01/24/2022

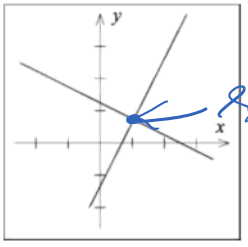
Monday, January 24, 2022 11:26 AM

M168_Notes_2.1

Chapter 2: Systems of Linear Equations and Matrices

2.1 Systems of Linear Equations: An Introduction

1. The two lines intersect.



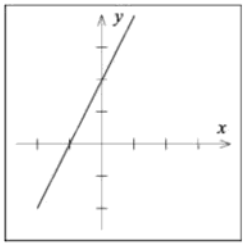
solution

Exactly one solution.

2. The two lines are the same.

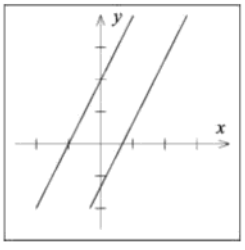
$2x + 3y = 7$

$4x + 6y = 14$



The entire line.

3. The two lines are parallel.



No solution

Example: Find the solution to the system of equations

$$\begin{aligned} x + 2y &= -1 \\ 2x + 4y &= -2 \end{aligned}$$

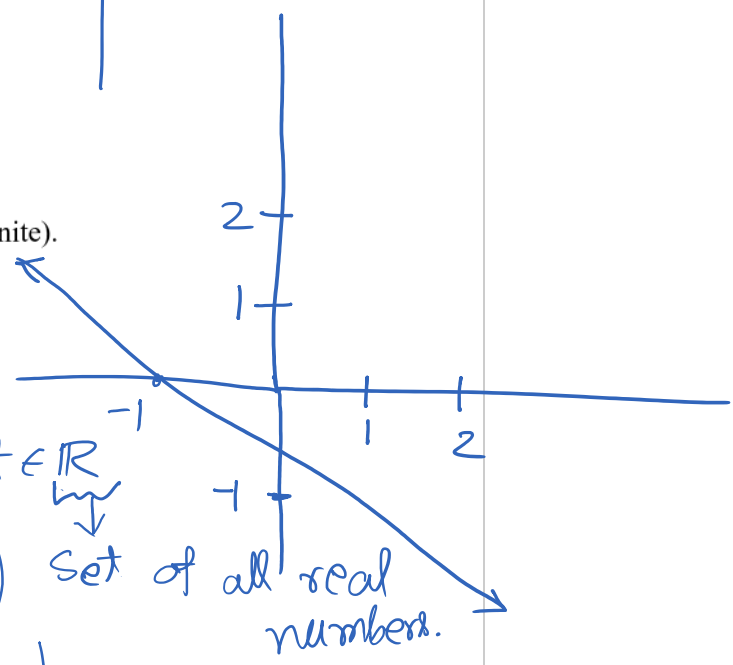
$\boxed{x = -1 - 2y}$

$2(-1-2y) + 4y = -2$

$\Rightarrow -2 - 4y + 4y = -2$

$\Rightarrow -2 = -2.$

$y = -\frac{1+x}{2} = -\frac{1}{2} - \frac{x}{2}.$



This system is called dependent.

The solution is a parametric solution (infinite).

We can consider y as a parameter,

$\boxed{(x, y) = (-2y-1, y)}$

$= (-2t-1, t), \; t \in \mathbb{R}$

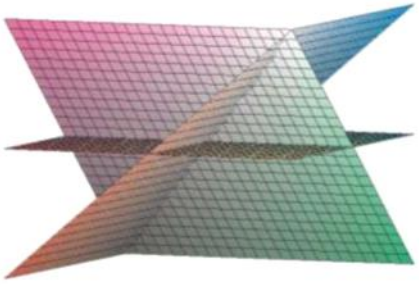
set of all real numbers.

Particular solutions:

$t = 0, \quad (-1, 0)$

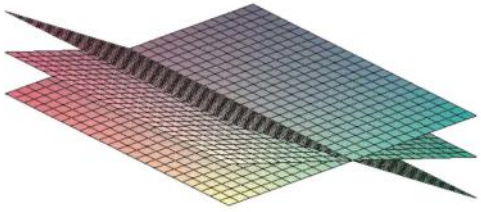
$t = 1, \quad (-3, 1)$

Example: Three Linear Equations with 3 Variables



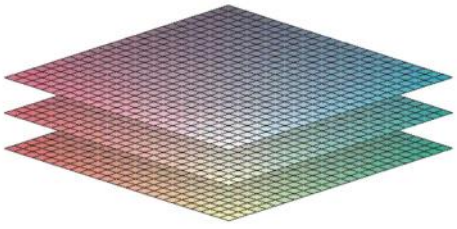
← Unique soln.

ONE SOLUTION



a line as solutions.

PARAMETRIC



NO SOLUTION

Formulating Linear Systems

An apartment complex is being developed that has small, large, and luxury apartments. The developer has decided that there will be a total of 192 apartments. The number of bigger apartments (large and luxury) will equal the number of small apartments. The number of small apartments will be three times the number of luxury apartments. How many apartments of each type will there be?

$x =$ # of small apartments

$y =$ # large apartments

$z =$ # lux apartments.

"The developer has decided that there will be a total of 192 apartments.

$$x + y + z = 192$$

"The number of bigger apartments (large and luxury) will equal the number of small apartments."

$$y + z = x$$

"The number of small apartments will be three times the number of luxury apartments."

$$3z = x$$

Zelda has $8,900 to invest. She decides to invest all of her money in three different funds. The PX Company costs $50 per share and pays dividends of $1.00 per share each year. The NY Company costs $90 per share and pays dividends of $2.00 per share each year. The LZ Company costs $40 per share and pays dividends of $1.60 per share per year. Zelda wants to invest half as much money in the LZ Company as in the NY Company and wants to earn $222 in dividends per year. How many shares of each company should Zelda buy to meet her goal?

$x =$ # shares of PX

$y =$ # of NY

$z =$ # of LZ

"The PX Company costs $50 per share and pays dividends of $1.00 per share each year." If you buy x shares in the PX Company,

How much does it cost? $ $50x$
How much do you earn in dividends? $ x

"The NY Company costs $90 per share and pays dividends of $2.00 per share each year." If you buy y shares in the NY Company,

How much does it cost? $ $90 \cdot y$
How much do you earn in dividends? $ $2y$

"The LZ Company costs $40 per share and pays $1.60 per share per year in dividends." If you buy z shares in the LZ Company,

How much does it cost? $ $40z$
How much do you earn in dividends? $ $1.6z$

Write an equation for the statement "Zelda has $8,900 to invest."

$$50x + 90y + 40z = 8900$$

Write an equation for the statement "Zelda wants to earn $222 in dividends per year."

$$1 \cdot x + 2 \cdot y + 1.6\, z = 222$$

"Zelda wants to invest half as much money in the LZ Company as in the NY Company."

$$90y = 2(40 \cdot z) \Rightarrow \boxed{40z = 45y}$$

Put it all together for your answer:

$$50x + 90y + 40z = 8900$$
$$x + 2y + 1.6z = 222$$
$$45y - 40z = 0$$